AstraCore CPU — ASIC Synthesis & Verification Report

AstraCore CPU is a custom 5-stage pipelined RISC-V style processor developed in Verilog as part of the AstraCore SoC platform. The processor implements forwarding logic, hazard detection, load-use stalling, branch flushing, JAL/JALR support, RTL verification, and ASIC-oriented synthesis using the SKY130 HD standard cell library.

1. CPU Standalone Performance Metrics

Metric	Value
Total Cycles	18
Instructions Executed	11
Stall Count	1
Flush Count	1
CPI	1.636364

Observations:
• Single load-use stall inserted correctly by Hazard Detection Unit
• Branch flush logic verified successfully
• Forwarding logic reduces RAW hazard penalties
• CPI overhead mainly caused by unavoidable pipeline hazards

2. ASIC Synthesis Summary

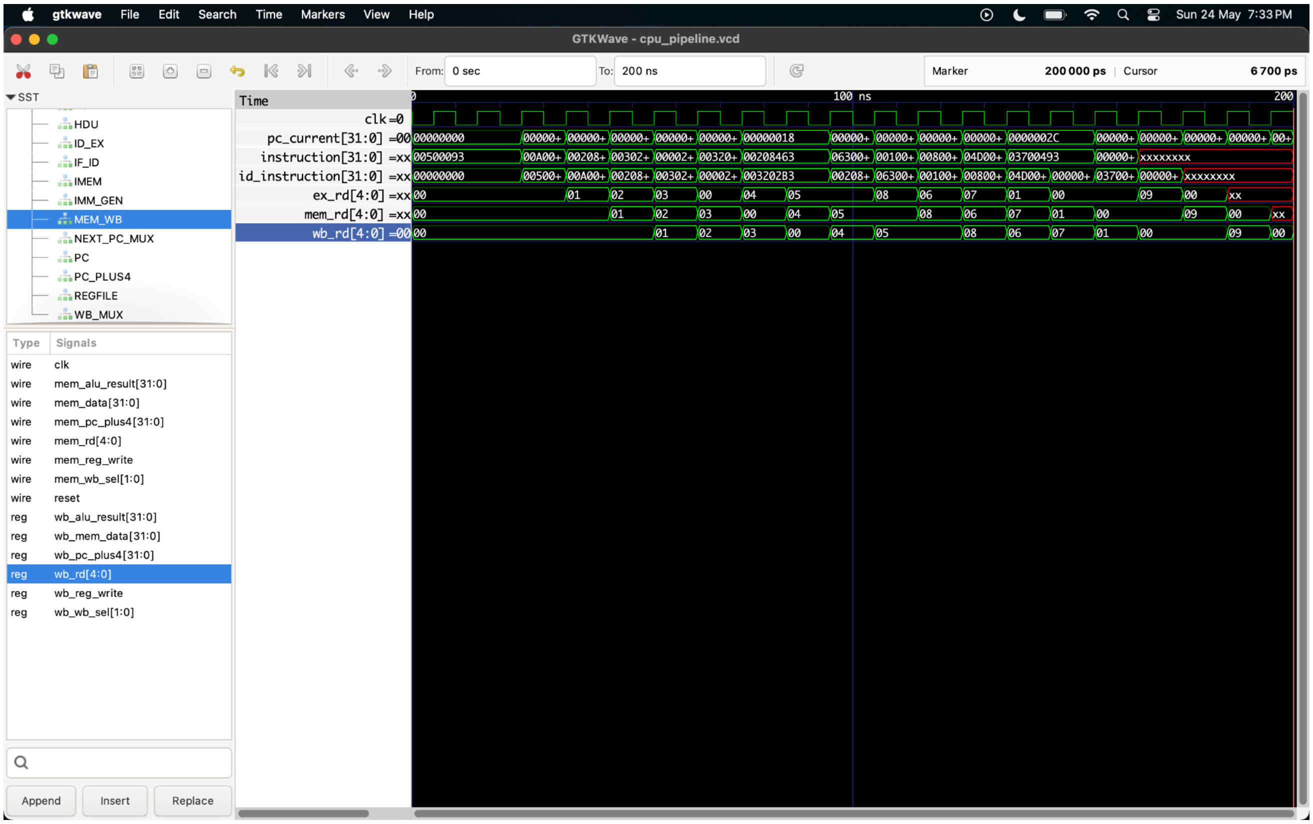
Metric	Value
Technology Library	SKY130 HD Standard Cell Library
Total Standard Cells	3317
Total Synthesized ASIC Area	16358.1888 μm^2
RTL Complexity Cells	36,307
Technology-Mapped Cells	19,725

3. Major Module Area Distribution

Module	Area (μm^2)
ALU	7046.7584
Branch Target Adder	948.4096
Comparator	724.4448
Next PC Mux	1072.2784
Forwarding Unit	252.7424
Hazard Detection Unit	130.1248

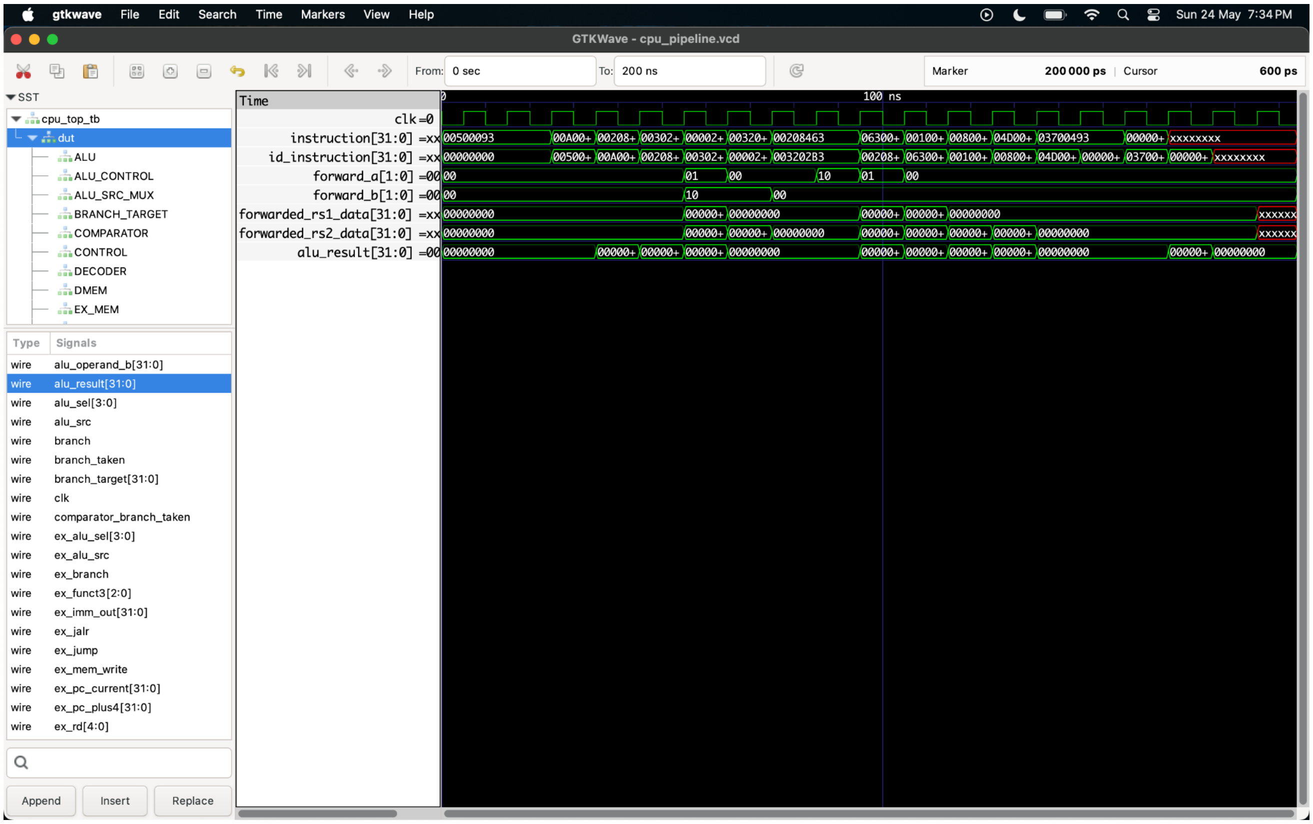
Writeback Mux	573.0496

Pipeline Register Propagation



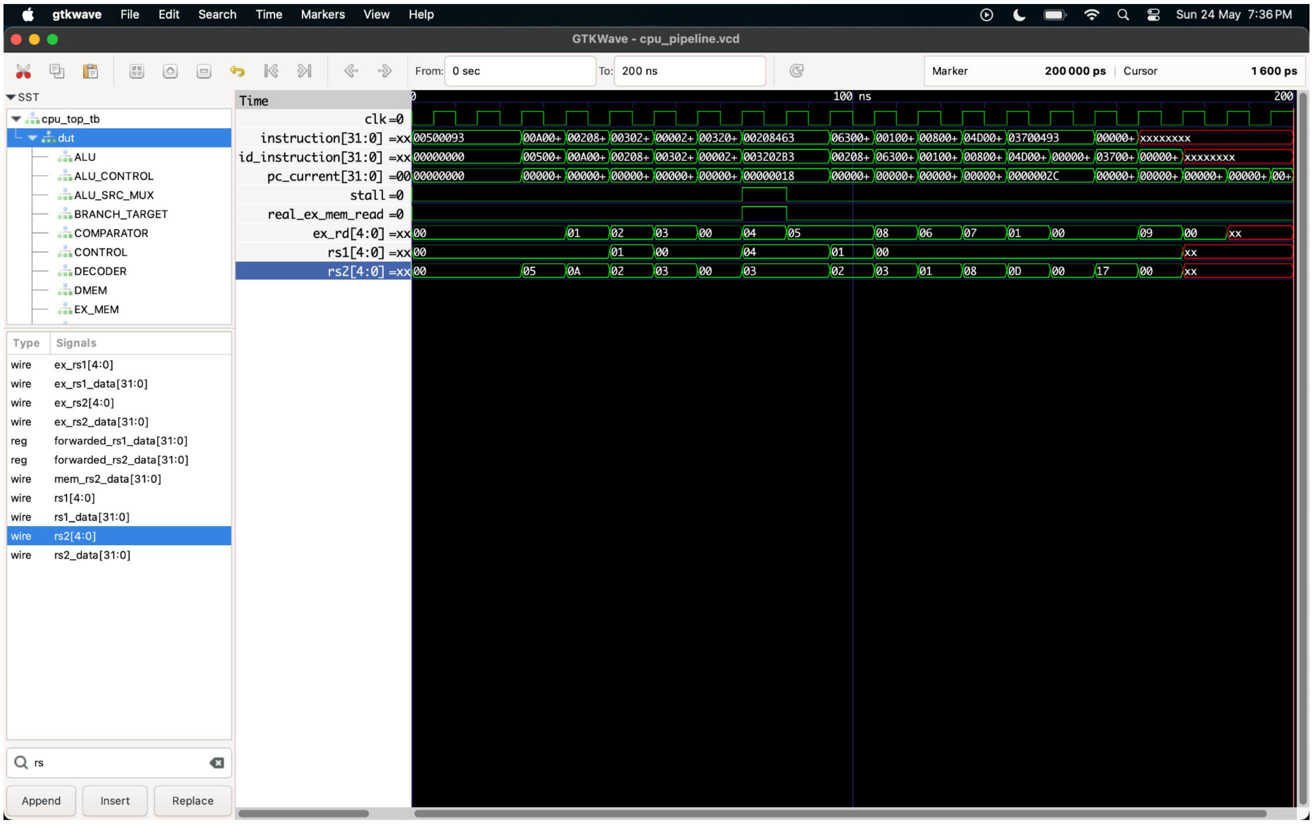
Waveform showing instruction propagation across IF/ID, ID/EX, EX/MEM, and MEM/WB pipeline stages. Register destination tracking confirms correct pipeline progression.

Forwarding Unit Verification



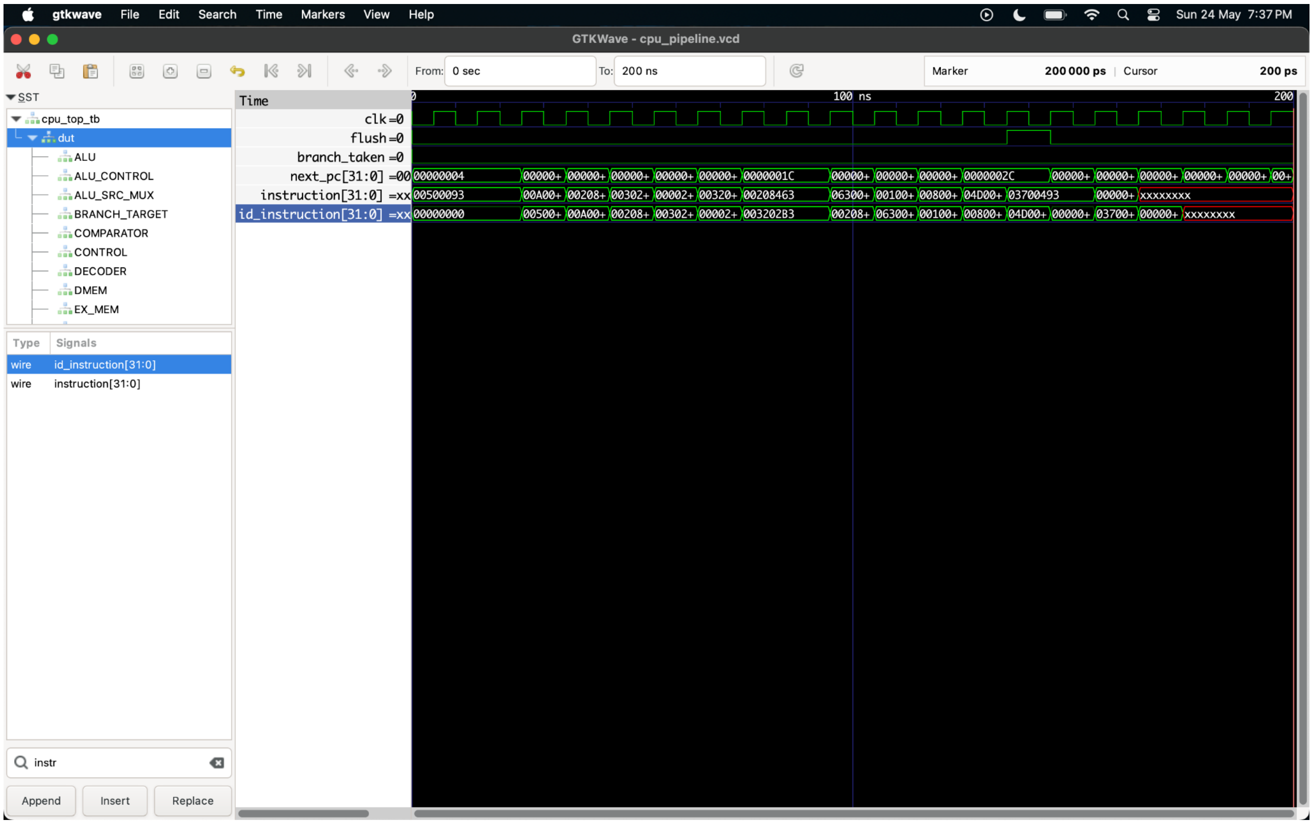
Waveform demonstrating forwarding logic activation through forward_a and forward_b control signals. RAW hazard mitigation verified successfully.

Hazard Detection and Stall Verification



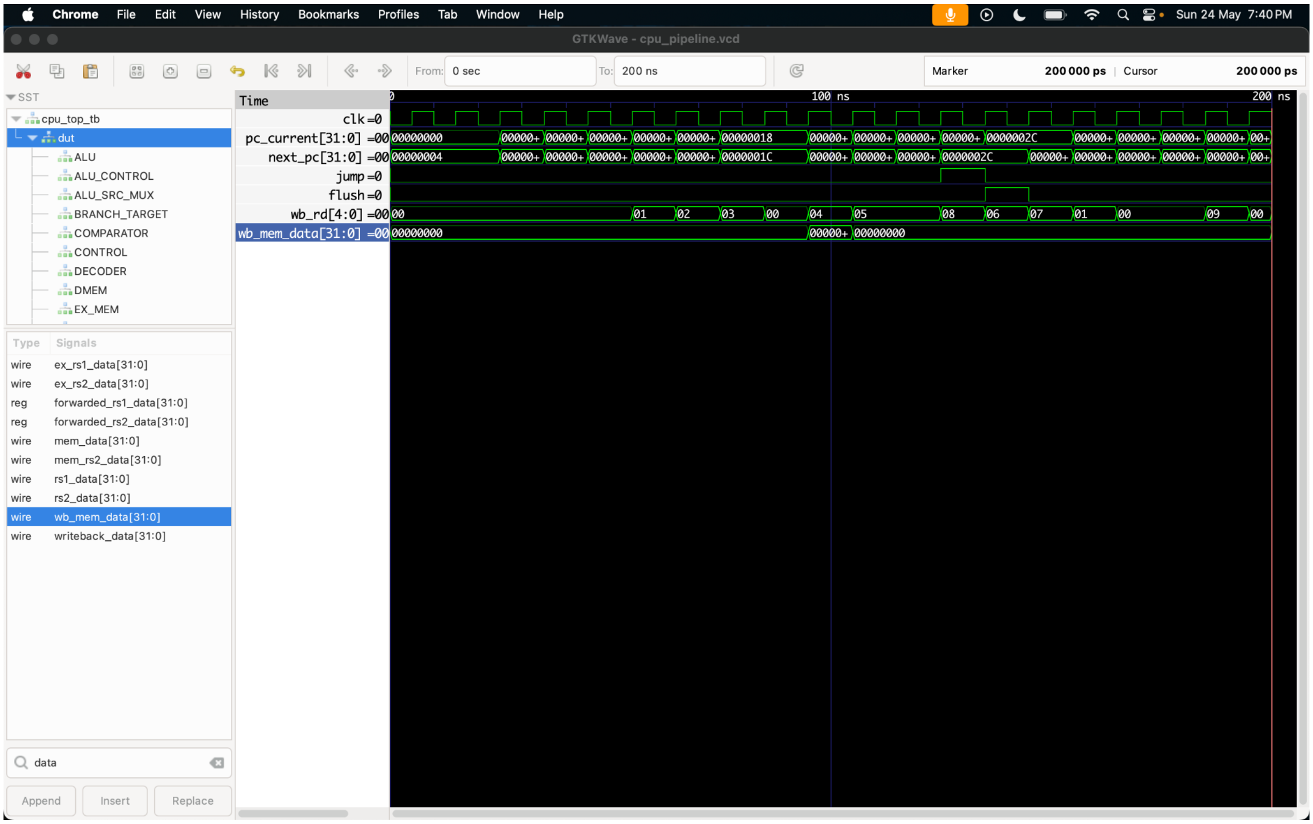
Waveform showing stall assertion during load-use dependency condition. Hazard detection logic correctly inserts pipeline stall cycle.

Branch Flush Verification



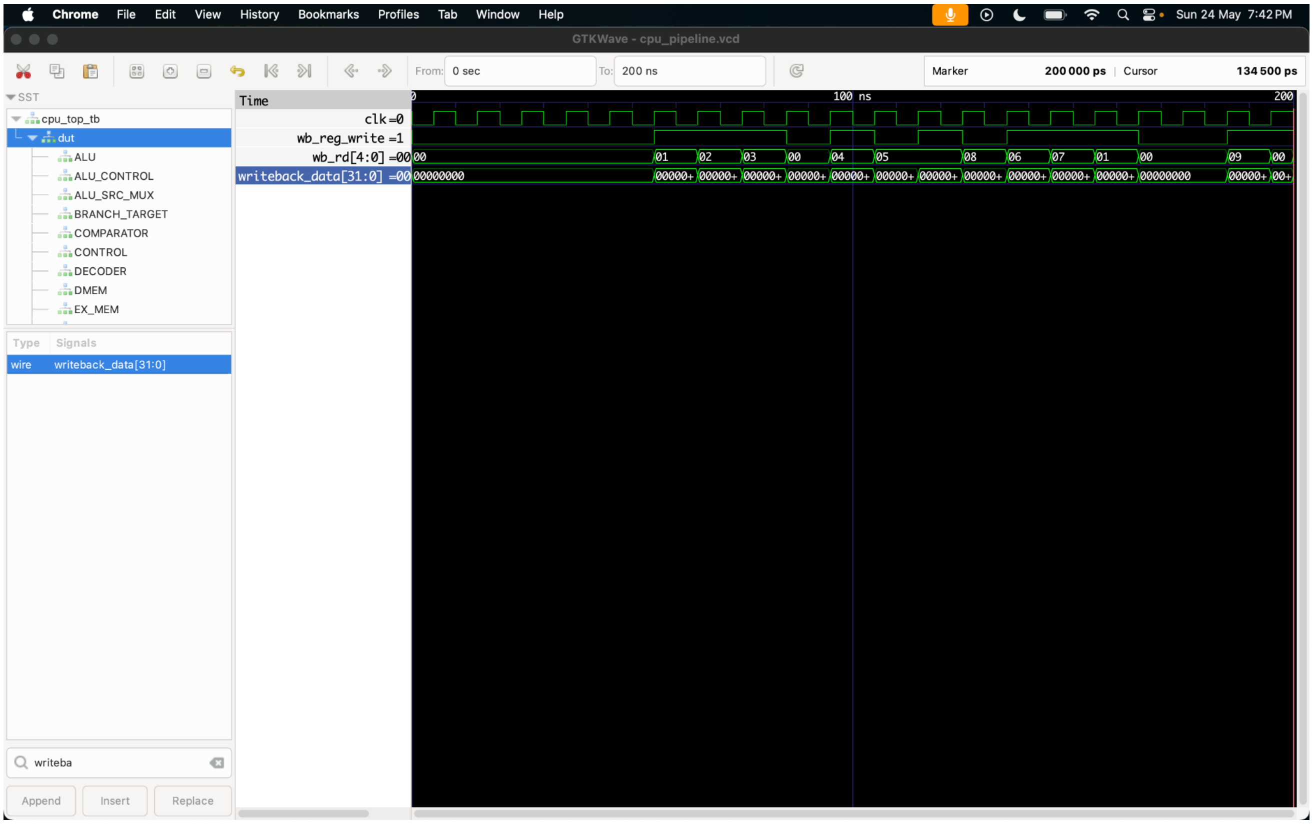
Waveform showing flush assertion during branch redirection. Incorrect speculative instruction removed successfully from pipeline.

Jump / Next PC Logic Verification



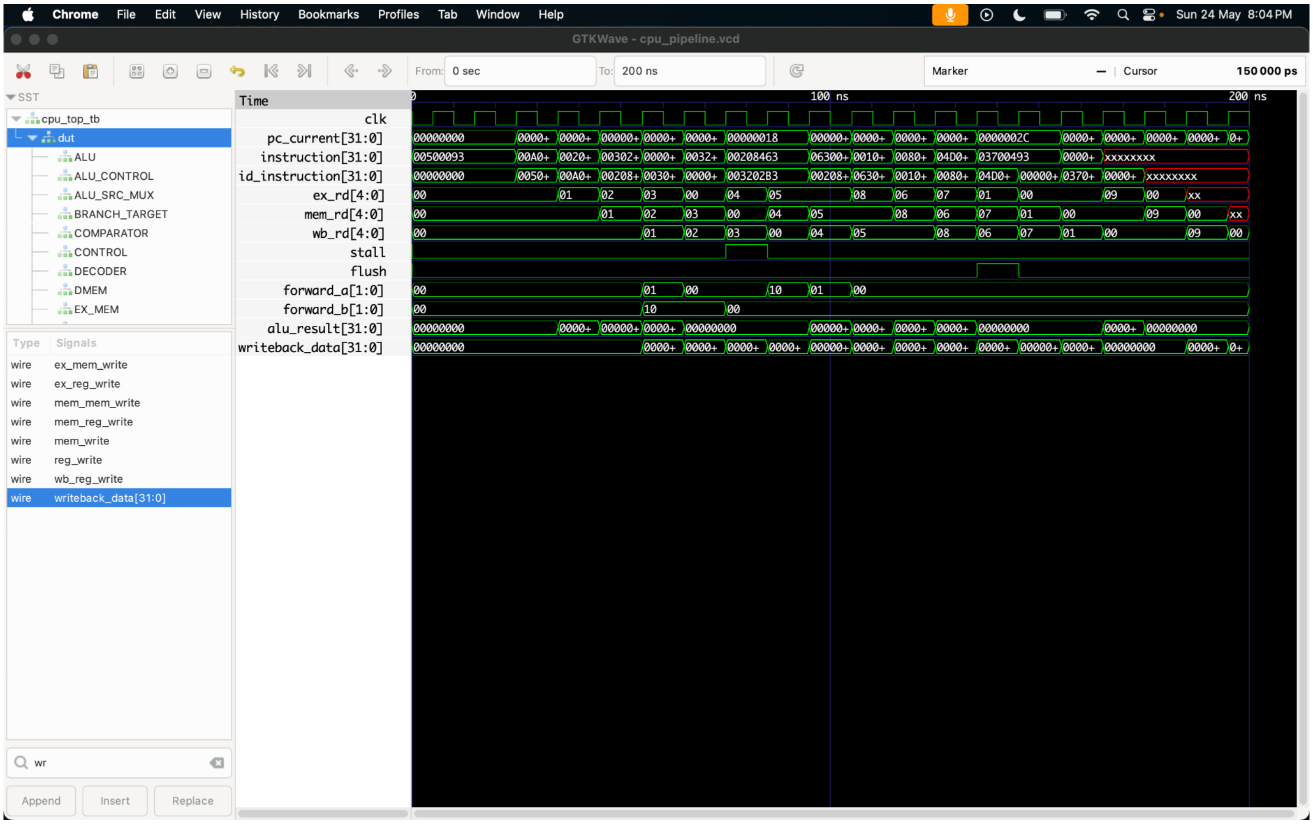
Waveform showing jump control activation and next PC redirection logic. PC update behavior verified successfully.

Writeback Data Path Verification



Waveform showing register writeback data propagation and destination register updates through WB stage.

Integrated Pipeline Verification



Combined waveform demonstrating forwarding, stalls, flushes, pipeline register flow, ALU results, and writeback operation together.

4. Implemented Features

• 5-stage pipelined CPU architecture
• Hazard Detection Unit
• Data Forwarding Unit
• Load-use stalling
• Branch flushing
• JAL / JALR support
• ALU datapath implementation
• Pipeline register implementation
• Register writeback logic
• Branch comparator and target logic
• Memory stage integration
• ASIC-oriented synthesis flow

5. Backend Status

Stage	Status
RTL Design	Completed
ASIC Synthesis	Completed
SKY130 Technology Mapping	Completed
Initial Backend Exploration	Completed
Full Routed STA	Deferred
CTS / Power Analysis	Deferred
OpenLane Physical Design	Planned

6. Current Limitations

1. Some sequential cells were not fully mapped into SKY130-specific sequential primitives.
2. Accurate timing closure requires full backend physical design flow.
3. Routed parasitic extraction and power estimation remain future work.
4. Full SoC-level integration is still under development.